

Abu Sufian Robin

CSE Graduate | Software Engineering, Applied ML, Research Projects



Date of Birth: 21/06/2002

📍 Dhaka, Bangladesh

✉️ abusufian02robin@gmail.com

☎️ +880 1407-993877

🔗 abusufianrobin.github.io • in abusufianrobin

🐙 abusufianrobin

Professional Summary

Recent CSE graduate from United International University with hands-on experience in software engineering, applied machine learning, and research-driven system development. Built full-stack applications, computer vision pipelines, and data-driven prototypes across healthcare, agriculture, fraud detection, and analytics domains. Experienced in Python, JavaScript, React, Node.js, Flask/Django, SQL, Git/GitHub, and experimental model evaluation. Seeking a Research Assistant – Software Engineering internship where I can contribute to software development, research implementation, testing, documentation, and product-oriented experimentation in a full-time on-site role.

Core Skills

Languages	Python, JavaScript, C, C++, Java, PHP, HTML/CSS
ML / AI	PyTorch, Scikit-learn, YOLO, Federated Learning, Homomorphic Encryption, Random Forest, Naive Bayes, KNN, Logistic Regression
Web / Backend	React.js, Node.js, Express.js, Django, Flask, Streamlit, Tailwind CSS, Bootstrap
Databases	MySQL, MongoDB, Google Firebase
Cloud & DevOps	AWS (Cloud Essentials Certified), Docker, Power BI, REST APIs, Figma
Soft Skills	Technical leadership, event management, cross-functional collaboration, English proficiency (US Embassy certified)

Experience

Intern, Software Team

Nov 2025 – Present

UIU Advanced Underwater Robotics & Automation (AURA) | Dhaka, Bangladesh

- Participated in the **MATE ROV Competition** as part of the Mariner team, focusing on underwater robotics systems.
- Developed underwater **object detection, classification, and recognition** pipelines using real-time cameras and computer vision models, including the **You Only Look Once (YOLO)** object detection model.

Public Relations & Event Coordinator

Aug 2023 – June 2024

UIU Robotics Club | Dhaka, Bangladesh

- Communicated with marketing representatives from food and technology companies to secure sponsorships for club events.
- Coordinated invitations for industry representatives to participate in seminars, workshops, and competitions organized by the club.
- Supported the organization of robotics and hardware competitions, enabling students to participate in hands-on learning experiences.

Key Projects

Credit Card Fraud Detection System | Python

Project Details

- Designed a privacy-preserving fraud detection system using homomorphic encryption and federated learning to enable secure decentralized model training without exposing sensitive financial data.
- Implemented deep learning models to detect fraudulent transactions, achieving **96%** detection accuracy while maintaining strong data privacy compliance.

AgroLand E-Commerce Platform | *JavaScript, Python*

Project Details

- Planned and developed a full-stack agricultural web application connecting farmers, agribusiness managers, and agricultural researchers.
- Integrated AI-powered features including a chatbot using **Meta LLaMA 3.1** and personalized crop recommendations using Random Forest and Logistic Regression, along with APIs for weather forecasting, marketplace services, and equipment/storehouse rental.

Multi-Dataset Machine Learning Prediction Suite | *Python*

Project Details

- Applied KNN, Naive Bayes, and Logistic Regression across six datasets (heart disease, IMDB ratings, sports/weather decisions, semantic image segmentation, Bangladesh climate data) covering binary and multiclass classification. Evaluated models using accuracy, precision, and recall.

Skin Burn Detection and Integration with Tiny SLM for Clinical Suggestions | *Computer Vision*

- Developed a multi-staged framework using **YOLOv8** to automate burn severity classification from wound images, achieving a competitive **mAP50 of 81.0%** and **85.0% recall** on a dataset of 15,216 images.
- Integrated a lightweight **Google Gemma-2B** model to analyze visual detections and generate contextual clinical suggestions, bridging the gap between computer vision and language-based medical reasoning.
- Designed the system for emergency and low-resource healthcare settings to reduce diagnostic delays and support medical professionals during initial patient examinations.

Super Store Sales Dashboard | *Power BI*

Project Details

- Developed a comprehensive Sales Data Analysis dashboard in Power BI to identify key revenue drivers and seasonal trends of a sales store.
- Implemented Time Series Analysis and Sales Forecasting models to project future performance with approximately **85% to 87%** accuracy, enabling data-driven inventory planning.
- Performed advanced data modeling and Data Analysis Expressions (DAX) computations to visualize Key Performance Indicators (KPIs), resulting in a **15%** improvement in trend identification efficiency.

Real-Time Weather Forecast Dashboard | *Power BI, Rest API*

Project Details

- Engineered a real-time Weather Analytics dashboard integrating live API data to track temperature, humidity, and Air Quality Index (AQI).
- Designed a predictive 7-day weather forecast interface featuring automated monitoring of environmental parameters like Wind Speed, Sunrise, and Sunset timings.
- Optimized data refresh schedules and visualization layouts to ensure seamless real-time monitoring of fluctuating weather patterns.

Education

United International University — *Dhaka*

October 2022 – March 2026

BS in Computer Science

- **CGPA:** 3.33/4.0 (Degree Verification)
- **Coursework:** Computer Architecture, Comparison of Various Learning Algorithms, Computational Theory.
- **Achievement:** Awarded a **25% Merit Scholarship** for outstanding academic performance in the **Fall 2022** and **Summer 2024** semesters.

Dr. Mahbubur Rahman Mollah College — *Dhaka*

June 2019 – February 2022

Higher Secondary School (HSC)

Group: Science

- **GPA:** 5.00/5.00 (HSC Certificate)
- **Board:** Dhaka
- **Achievement:** Awarded a full-time **BGMEA scholarship** in recognition of achieving GPA 5.00 in the Higher Secondary Certificate (HSC) Examination.

Research Works

[Empirical Studies on Green Computing and Edge AI in Agriculture]

[1] A SLM-Driven Multi-Modal Edge Computing Framework for Soil and Crop Management

2025 – Present

- Overview*
- Designed a lightweight edge-based framework integrating SLMs, RAG, and knowledge graphs for precision agriculture.
 - Enabled real-time multi-modal analysis using soil metrics, environmental data, and crop imagery.
 - Delivered on-device recommendations for fertilization, irrigation, and disease detection in low-connectivity settings.
- Outcome* Achieved **94%** disease detection accuracy and **87.5%** F1-score, with only **4%** hallucination, demonstrating reliable real-time recommendations.

[Empirical Studies on Multi-Omics Data Integration in Genomics]

[2] A Novel Approach for Combining Single-Cell ATAC and RNA Sequencing for Supervised Cell Annotation

2025 – Present

- Overview*
- Proposed a novel framework integrating single-cell ATAC-seq and RNA-seq for improved cell annotation.
 - Applied dimensionality reduction and automated gene-based annotation for multi-omics data fusion.
 - Enhanced cell identity and state prediction through advanced integration and balancing techniques.
- Outcome* Achieved **93.13%** F1-score on PBMC and **99.90%** on AD neuronal datasets, showing significant improvement (up to **+14.7%**) over baseline methods.

Certifications & Professional Development

- | | |
|--|--|
| i. MS EXCEL <i>Certificate</i> | vii. Facebook Marketing <i>Certificate</i> |
| ii. MS PowerPoint <i>Certificate</i> | viii. Leadership <i>Certificate</i> |
| iii. Digital Security <i>Certificate</i> | ix. Red Hat System Administrator <i>Certificate</i> |
| iv. Cisco Networking <i>Certificate</i> | x. AWS Cloud Technical Essentials <i>Certificate</i> |
| v. IoT – Internet of Things <i>Certificate</i> | xi. Full Stack Development <i>Certificate</i> |
| vi. AI/ML <i>Certificate</i> | |

Language Proficiencies

- English *Certificate* from *English Conversation Club, EMK Center, USA Embassy, Dhaka*

Additional Activities

- | | |
|--|---|
| i. Photography <i>Certificate</i> | iii. Fact-Checking <i>Certificate News Media, USAID</i> |
| ii. Campus Ambassador <i>Certificate</i> | iv. Volunteer Activities at <i>UIU Robotics Club</i> . |

References

Dr. Riasat Azim

Assistant Professor, Department of CSE
United International University, Bangladesh
Email: riasat@cse.uiu.ac.bd
Tel: +880 1796-619200

Dr. Md. Motaharul Islam

Professor, Department of CSE
United International University, Bangladesh
Email: motaharul@cse.uiu.ac.bd
Tel: +880 1712-644837